

PAPER

Single Cell integration via Linear Assignment Problem

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Abstract

Single Cell RNA-seq data integration from multiple batches is a crucial step in experimental analysis with multiple competing methods being the status quo especially considering the growing distance of batches being considered i.e. going from same tissue to cross species integration.

We propose a method based on the Linear Assignment problem where batches are pairwise globally matched. We investigate a cutoff based on distance percentile that balances between competing goals in integration: cells of the same type should stick together versus the obvious degenerate solution where all batches are disconnected. We consider established competitor methods for comparison and consider a diverse set of quality measuring techniques. We collected 26 annotated datasets for our analysis, 11 of which are time series datasets that represent a more difficult integration task. Low number of datasets is still an issue with contemporary single cell works.

Key words: scRNA-seq Single Cell, Linear Assignment Problem, Data Integration

Introduction

Single-cell RNA sequencing (scRNA-seq) has revolutionized our understanding of cellular diversity by enabling gene expression profiling at single-cell resolution. A critical challenge in harnessing this technology is the integration of data from multiple experimental batches, which exhibit systematic variations [17]. As data integration ambitions grow from small-scale studies to large atlases or the drive to integrate time series data, the sources of variance between batches also increase. Effective data integration is thus paramount for drawing robust insights from large-scale scRNA-seq studies.

In the ‘easy’ case where batches are replicates of the same tissue under similar conditions, we can

Time series is especially difficult as

Integrating time-series data presents a distinct “partial alignment” challenge. In temporal datasets, the “batch” label is often confounded with biological progression (e.g., developmental time). Standard greedy algorithms, which seek to maximize mixing, risk “over-integrating” the data by forcing distinct developmental stages (e.g., Day 0 and Day 10) to merge, thereby crushing the biological trajectory [8]. While methods like Scanorama have explicitly targeted this domain by attempting to stitch datasets together via overlapping edges [10], and trajectory inference tools often rely on batch-corrected inputs [33], the lack of a global disconnection mechanism often results in distorted manifolds.

Current integration strategies generally fall into three categories: linear regression models, dimensionality reduction techniques, and graph-based matchings. Linear methods, such as ComBat [11], model batch effects as additive noise but often fail to account for non-linear biological variations. Dimensionality reduction methods, such as Harmony [13] and Scanorama [10], project datasets into a shared embedding space, effectively aligning centroids or mutual nearest neighbors. Finally, graph-based approaches like BBKNN [27] and MNN [9] construct a joint neighborhood graph by identifying local links between batches. While effective in standard scenarios where batches share high cell-type overlap (e.g., replicates), these methods frequently struggle with the more complex task of longitudinal integration found in time-series experiments.

also mention scvi as a deep learning method. doesn't seem overwhelming in scbi, docs say it needs gpu, and is not in scanpy :) ... other methods that do well in the scbi paper require labels

In this work, we propose **Scalp**, a method that shifts the integration paradigm from local greedy matching to global optimization. We formulate batch alignment as a Linear Assignment Problem (LAP) [14], which acts as a discrete realization of Optimal Transport [30]. Unlike greedy approaches that optimize cell pairs in isolation, our method enforces a global constraint where the total distance across the system is minimized. Crucially, we introduce a distance percentile cutoff that allows the

algorithm to distinguish between "true" biological neighbors (to be integrated) and distinct biological states (to remain separate). This allows Scalp to robustly handle the varying degrees of overlap inherent in time-series and cross-species data.

We validate our approach on a diverse benchmark of 26 datasets, including standard integration tasks and 11 challenging time-series datasets. We demonstrate that by balancing global matching with hubness-corrected distance metrics for graph construction, Scalp achieves a superior trade-off between batch mixing and the preservation of biological trajectories.

Methods

Graph-based Integration of Single-cell Data We developed a graph-based approach for integrating multiple single-cell datasets that preserves both the local structure within each dataset and establishes meaningful connections across datasets. Our method builds upon previous work in graph-based representation learning for single-cell data [37, 20, 10] while addressing specific challenges in multi-dataset integration.

Problem Formulation: Given m single-cell datasets $\mathcal{X} = X_1, X_2, \dots, X_m$, where each dataset $X_i \in \mathbb{R}^{n_i \times d_i}$ contains n_i cells measured across d_i features, our goal is to construct a unified cell-cell similarity graph $G = (V, E)$. In this graph, vertices V represent all cells across datasets, edges E capture meaningful relationships between cells. The resulting graph can then be used for downstream tasks such as dimensionality reduction, clustering, and visualization.

Within-dataset Graph Construction: For each dataset X_i , we construct a weighted adjacency matrix G_{ii} representing its k -nearest neighbor (k-NN) graph. First, we calculate a matrix of pairwise Euclidean distances, D_i . Then we apply hubness correction to adjust for issues stemming from high dimensionality [29]. Finally, we produce the adjacency matrix directly from the corrected distances.

Cross-dataset Integration: To establish connections between datasets X_i and X_j , identified by G_{ij} , where $i \neq j$, we compute the euclidean pairwise distance matrix and apply hubness-correction to obtain D_{ij} . We then identify optimal cell-to-cell matchings using the linear assignment problem (Hungarian algorithm) [14, 22], which produces a matrix of "marked" cells such that each cell is matched to a cell from the other batch under the constraint that the sum of distances is minimized: $H_{ij} = \text{LinearAssignment}(D_{ij})$. Crucially H_{ij} can be further reduced. Each marked item is associated with a distance in D_{ij} (i.e. non-zero elements in the Hadamard product $H_{ij} \odot D_{ij}$) we can remove the largest distances based on a percentile cutoff and remove them from H_{ij} . Finally G_{ij} is constructed via the matrix multiplication $H_{ij}G_{jj}$.

Integrated Graph Assembly:

At last we can combine all parts to obtain G . With G as context, we can interpret the previous matrix multiplication: copying rows from G_{jj} to G_{ij} means introducing edges between instances of X_i and to the neighbors of its (Hungarian induced) matches in X_j (since we copy its row from the nearest neighbor graph matrix).

The resulting integrated graph preserves local structure within datasets while establishing biologically meaningful connections across datasets, enabling joint analysis and visualization of heterogeneous single-cell data collections. In this work, we utilize UMAP[19] where required to obtain low-dimensional embeddings from the integrated graph; in the definition of G we omitted edge weights, considering the graph construction method, these could be conserved easily, however transforming G into a boolean matrix halves the execution time with minimal impact on quality.

Implementation details:As a hubness correction method we use Cross-domain similarity local scaling [5], which performs similar to other local scaling techniques. Our implementation uses SciPy's sparse matrix representations for memory efficiency. The function accepts input in multiple formats, including AnnData objects [37] common in single-cell analysis.

Data

Our study utilizes a diverse collection of publicly available single-cell RNA-sequencing (scRNA-seq) datasets to evaluate our method. These datasets are divided into two primary categories based on the analytical challenge they represent: benchmark datasets (15) for data integration and time-series datasets (10) for a more difficult challenge. We use the labels provided by the datasets. After preprocessing the average dataset contains 4607 cells in 7.7 batches across 15.8 cell types.¹

Batch Datasets To assess the performance of batch correction and data integration, we first employed datasets from the scib benchmark, a comprehensive framework for evaluating single-cell integration methods [17]. We selected four tasks representing common challenges: a Pancreas dataset to test integration across different technologies [32, 17]; a Lung dataset with significant inter-donor variation [17]; a Human Immune dataset with highly similar cell types from different tissues; and a cross-species Human-Mouse Immune dataset to evaluate alignment across large evolutionary distances [17].

To further test our method in complex disease contexts, we incorporated an additional collection of 11 scRNA-seq datasets from cancer studies. This collection, originally assembled and curated for a comprehensive cancer benchmark [18], provides a robust testbed for integration by spanning multiple cancer types, including breast cancer [1, 2, 31], pancreatic ductal adenocarcinoma [7, 25], melanoma [4, 12], lung cancer [16, 28], and colorectal cancer [23, 38].

Time-Series Datasets For evaluating performance on tasks related to developmental dynamics and trajectory inference, we assembled a collection of datasets with well-defined temporal or differentiation processes. This collection includes datasets mapping human hematopoietic differentiation [36], fetal lung development [34], pancreatic endocrinogenesis [3], whole-organism regeneration in planaria [26], and nervous system development in the mouse cortex and cerebellum [33].

¹ Note that the large-scale mouse brain dataset described in the original benchmark was excluded, as it was not available in the provided repository.

Finally, we included four canonical datasets commonly used for trajectory inference benchmarking, made available via the CellRank toolkit [15]. These datasets profile mouse bone marrow differentiation [24], lung epithelium development [35], pancreas endocrinogenesis [3], and iPSC reprogramming [21], providing a standard reference for performance on established biological processes.

Data Preprocessing

The datasets, consisting of multiple batches each, underwent a standardized preprocessing and feature selection workflow. Initially, each batch was processed in isolation. We performed a quality control step by filtering out genes detected in fewer than three cells. The expression data was then normalized to 10,000 reads per cell and subsequently log-transformed with a pseudocount of one (log1p transformation) to stabilize variance. Following this standard procedure, gene variability was estimated for each batch using Cell Ranger [39]. To select a robust set of highly variable genes (HVGs) across the entire dataset, a binary search algorithm was used to find a single variability cutoff such that the union of selected genes is (just below) 2000 genes. The Batches were then combined, removing genes below the threshold and cells such that the batches are of the same size. Finally, we applied Principal Component Analysis (PCA) to this filtered data to further reduce its dimensionality to the top 40 principal components. Only where we report results for specific cell types (i.e. some figures), we removed infrequent cell types entirely to preserve legibility.

Results

We evaluate our method on 26 Batch and timeseries benchmark datasets. Since time series data may exhibit a cell type shift, we consider it more challenging. We characterized the datasets thoroughly in section 2.1. We compare our method to established and commonly used single cell integration methods: Harmony, Scanorama, BBKNN and ComBat. We consider previous metrics (SCIB and KNI) and our own for a quantitative comparison. These evaluate how well batches are mixed and how pure local neighborhoods are in terms of cell type. In a qualitative analysis we demonstrate differences to our closest competitor. Our method provides a distance cutoff parameter that regulates batch merging and cell type preservation. Its effects are analyzed via an example. We complete our results with a brief look at runtimes.

Current ScRNA-Seq Integration Ecosystem

The integration of scRNA-seq datasets is addressed by several established methods, which are implemented within the popular Python-based Scanpy framework [37]. For our comparative analysis, we focus on four prominent algorithms: **BBKNN** [27], **ComBat** [11], **Harmony** [13], and **Scanorama** [10]. Since the mutual nearest neighbors (MNN) method [9] is falling out of favour as evidenced by its lack of maintainance in the current scRNA-Seq ecosystem, we exclude it from our study.

These selected methods operate under different assumptions and employ distinct correction mechanisms, ranging from graph-based batch-balancing to linear

model adjustment and mutual nearest-neighbor alignment. A common theme among them is their reliance on **local optimization** strategies, which correct batch effects by aligning cells within neighborhoods or clusters. In contrast, our proposed method, **SCALP**, performs **global alignment** of datasets. We provide a concise overview of each competitor’s approach.

BBKNN [27] constructs a batch-balanced k-nearest neighbour graph by computing neighbours within each batch and combining them, thereby preserving local structure while aligning neighbourhoods across batches. ComBat [11] models batch effects as additive and multiplicative shifts, using an empirical Bayes framework to estimate and remove them given batch covariates. Scanorama [10] identifies mutual nearest neighbours across datasets in a low-dimensional space, computes linear transformations to align them, and incrementally merges datasets to produce a panorama that integrates shared cell populations while retaining biological variation. Harmony [13] initializes distances between cells in batches by their distance in shared space, then alternates between soft clustering and linearly adjusting each batch. The adjustment moves each batch along each dimension, such that the cells maximize their probability to belong to a cluster.

Quantitative Evaluation

We explore multiple frameworks on scoring for this task, as several have been established and introduce our own method. We will see that our method delivers solid performance on all metrics.

Our method provides a crucial parameter that controls the connectedness between batches. We explore the impact of this parameter on overall performance and how well it balances batch mixing and cell type cohesion; we call these the batch and the label measure.

k-NN based /evaluation

To determine the quality of batch integration there are two goals: cells of the same type should to group together and cells from different batches should be in close proximity to make sure an actual integration happened. Both can be measured by a k nearest neighbor model. We use 5 fold crossvalidation on 5 neighbors. For the cell type we obtain our score directly from this procedure. However, for the batch id, we use the complement of the score, i.e. $1 - x$, not being able to guess the batch from its neighbors means that the batch-ids of cells are mixed and a high score indicates good performance.

Method	Label	Batch	(g)Mean
Scalp: 0.2	0.64	0.42	0.52
Scalp: 0.4	0.56	0.64	0.60
Scalp: 0.7	0.42	0.83	0.59
BBKNN	0.61	0.46	0.53
ComBat	0.47	0.61	0.53
Harmony	0.62	0.47	0.54
Scanorama	0.57	0.50	0.53

KNI Metric

A simple scoring technique was proposed Diaz-Mejia et al.[6] that combines batch and label scoring in the following way: If the a cells neighbors are from different batches, make a nearest neighbor prediction on the label, if the

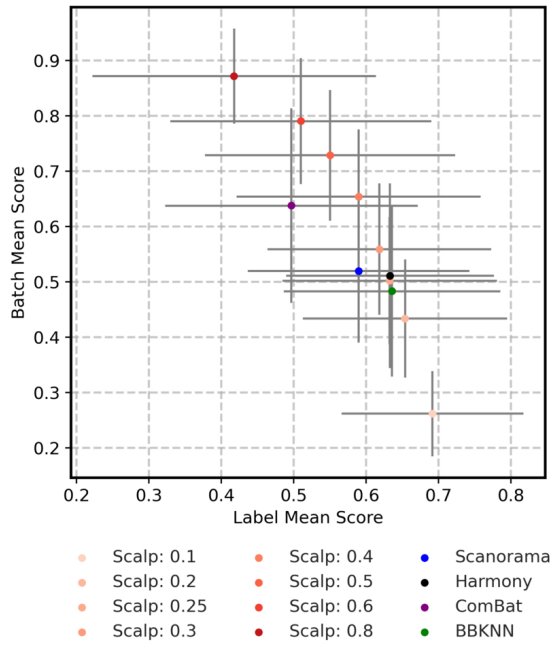


Figure 1. performance in two dimensions for various methods. for each of our datasets, we calculated the batch and label scores, then reduced the data to mean and variance on both metrics. the batch metric indicates the batch-label-variance in local neighborhoods while the label metric indicates label-uniformity in local neighborhoods, i.e. higher is better for both metrics. interestingly, different settings for our main parameter results different points along the pareto front of solutions.

cell-type is correctly identified, the cell gets a score of 1, otherwise 0. The final score is the average score for all cells. We calculated this score for all datasets and report the mean. Since we do not exclude smaller cell populations, we considered 10 neighbors for the calculation of the score, while keeping other defaults. The evaluations by Diaz-Mejita et al. suggest that this is reasonable. Since our method produces an adjacency graph, we provide the hubness adjusted distance matrix.

Our competitors scores are plausible Harmony:0.53, Scanorama: 0.51, BBKNN: 0.56, and ComBat: 0.10. Considering the results of Fig. 2 and 1, where combat performs slightly worse, but SCIB disagrees (table ??). For our method, the results range from a score of 0.21(Thresh=0.1) to 0.59(Thresh=0.9). Considering Fig 6 and 5 this shows a clear issue with the metric. a very strict cutoff threshold disjoints the clusters in an eronious way, while KNI is offering a high score.

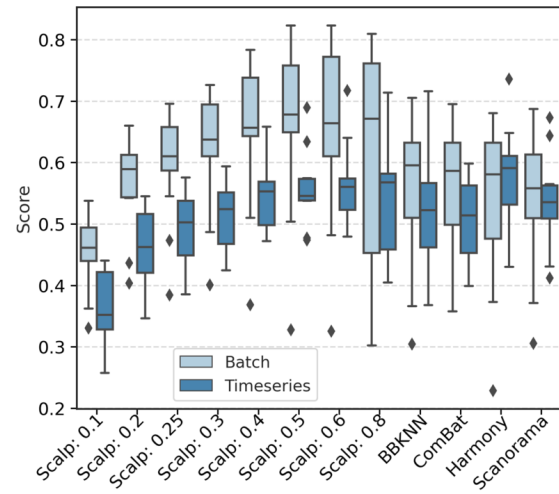


Figure 2. comparing methods across the dataset types. Our datasets consist of regular batch and in addition time series datasets. Here we examine the difference between the two sets. the batch metric indicates the batch-label-variance in local neighborhoods while the label metric indicates label-uniformity in local neighborhoods, i.e. higher is better for both metrics. time series datasets slices differ not only by batch but also by time of cell extraction. THus they are more difficult to integrate.

	0
Harmony	0.53
Scanorama	0.51
BBKNN	0.56
ComBat	0.10
Scalp: 0.1	0.21
Scalp: 0.15	0.29
Scalp: 0.2	0.35
Scalp: 0.25	0.41
Scalp: 0.3	0.46
Scalp: 0.35	0.50
Scalp: 0.4	0.53
Scalp: 0.45	0.55
Scalp: 0.5	0.57
Scalp: 0.55	0.58
Scalp: 0.6	0.58
Scalp: 0.65	0.59
Scalp: 0.7	0.59
Scalp: 0.75	0.59
Scalp: 0.8	0.59
Scalp: 0.85	0.58
Scalp: 0.9	0.58
Scalp: 0.95	0.57

SCIB Metric

A benchmarking reievew work [17] proposed to employ a host of metrics to determine the quality of integration, These can be divided into batch and label centric scores as well.

The label centered metrics are Adjusted Rand Index, Cell cycle conservation score, Cell-type LISI (cLISI) score, Highly variable gene overlap, Isolated label score ASW, Isolated label score F1, Normalized mutual information, Average silhouette width (ASW) and Trajectory conservation score. The batch focused metrics are Graph Connectivity Integration LISI (iLISI) score kBET score

Principal component regression score Batch ASW. These range from 0 to 1. We calculated the average for batch and label focused metrics and report the results in table ??.

Method	Label	Batch	(g)Mean
Scalp: 0.2	0.41	0.90	0.61
Scalp: 0.4	0.38	0.90	0.58
Scalp: 0.7	0.30	0.86	0.51
BBKNN	0.39	0.75	0.54
ComBat	0.37	0.87	0.57
Harmony	0.36	0.81	0.54
Scanorama	0.37	0.71	0.51

Table 1. bla

print all 4 full tables to the appendix, (with the /) reduced value versions with geomean scores goes to the paper for scib and ours... (check again for nice values, and select some wehre we can] actually make a case in the scib scoring)

for the largest datasets i can subsample multiple times... to show that there is not too much of adifference..

these tables, (there is also the one with our batch/label scores). report geomean and rank rename the data columns of these tables :)

Qualitative Evaluation

in figure 4 and 3 we see the comparison to according to quantitative analysis our fiercest competitor bbknn. Both datasets were preprocessed the same way. For visibility, we restricted the datasets to contain the largest clusters only.

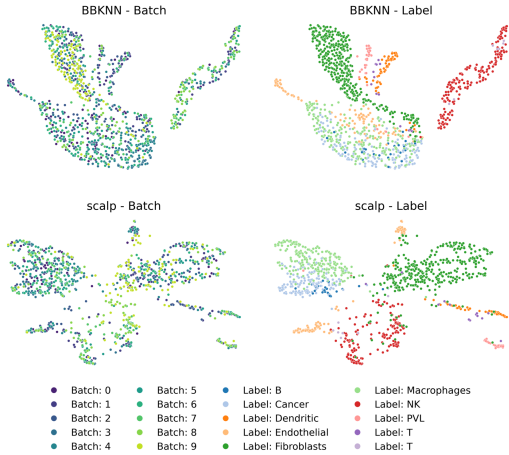


Figure 3. comparing bbknn and scalp on a batch data set. we observe umap embeddings for the two methods, colored according to batch and celltype labels each, resulting in four plots. Although both method exhibit similar failure modes, scalp is preferred. Both methods split endothelial cells in to two clusters. A fraction of Fibroblast cells is separated in bbknn while scalp distributes some instances. However in scalp cd4+T and dendritic as well as B, cancer and Macrophages are clearly more distinct.

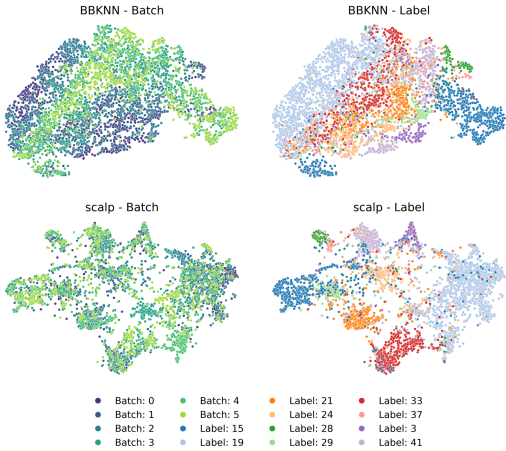


Figure 4. compare bbknn to scalp on a time series dataset. we observe umap embeddings for the two methods, colored according to batch and celltype labels each, resulting in four plots. The batch overview indicates that earlier and later batches are not well mixed when employing bbknn. A look at the labels reveals that indeed cluster 29 and 24 are victims of this split, while scalp sorts these out more coherently. In general, scalp gets some single instances wrong, however the global clustering makes sense compared to bbknn's solution.

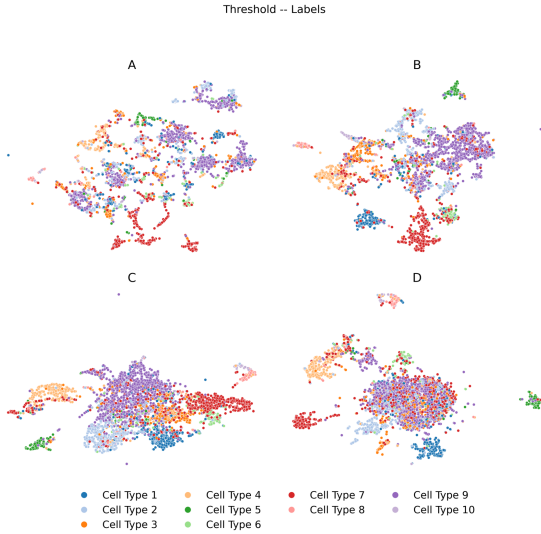


Figure 5. Time Series Data with varying cutoff threshold - showing Cell Types

The plot shows cutoffs at 10%(A), 40%(B), 70%*(C), and 90%(D). (this should be the 509? data)

runtime acknowledgements

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also the scanorama paper has an open access version. it should be more prominently placed

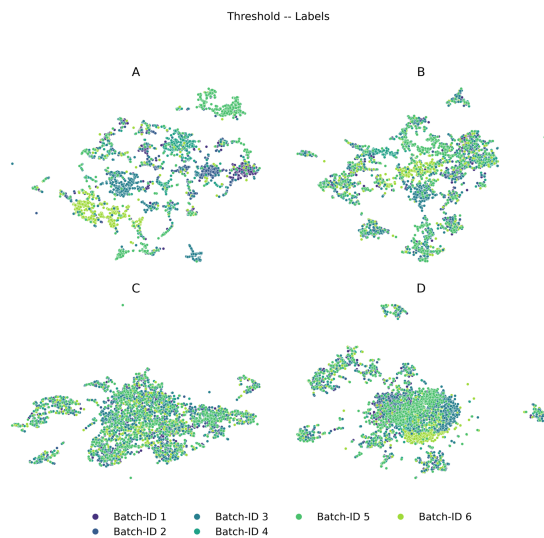


Figure 6. Highlighting Batches (time slices) for varying cutoff threshold

The plots show cutoffs at 10%(A), 40%(B), 70%*(C), and 90%(D). 3(this should be the 509? data)

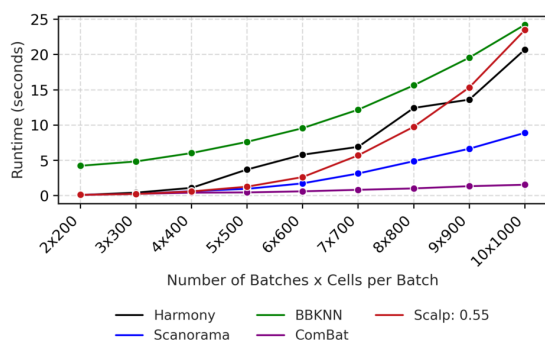


Figure 7. runtime comparison

Competing interests

No competing interest is declared.

Author contributions statement

Must include all authors, identified by initials, for example: S.R. and D.A. conceived the experiment(s), S.R. conducted the experiment(s), S.R. and D.A. analysed the results. S.R. and D.A. wrote and reviewed the manuscript.

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Method	Label	Batch	Geomean
Scalp: 0.1	0.68 /0.62/0.72	0.25/0.21/0.28	0.41/0.36/0.45
Scalp: 0.15	0.66/0.60/0.70	0.34/0.29/0.38	0.47/0.42/0.52
Scalp: 0.2	0.64/0.58/0.69	0.42/0.35/0.47	0.52/0.45/0.57
Scalp: 0.25	0.61/0.56/0.65	0.49/0.42/0.54	0.55/0.48/0.59
Scalp: 0.3	0.60/0.54/0.64	0.54/0.47/0.60	0.57/0.51/0.62
Scalp: 0.35	0.57/0.53/0.61	0.59/0.52/0.65	0.58/0.52/0.63
Scalp: 0.4	0.56/0.51/0.60	0.64/0.57/0.70	0.60/0.54/0.65
Scalp: 0.45	0.54/0.49/0.57	0.68/0.61/0.74	0.60/0.55/0.65
Scalp: 0.5	0.52/0.48/0.55	0.72/0.65/0.78	0.61 /0.56/0.65
Scalp: 0.55	0.50/0.46/0.52	0.75/0.68/0.80	0.61 /0.56/0.65
Scalp: 0.6	0.47/0.44/0.49	0.78/0.72/0.83	0.61 /0.56/0.64
Scalp: 0.65	0.44/0.42/0.46	0.81/0.74/0.86	0.60/0.56/0.63
Scalp: 0.7	0.42/0.40/0.43	0.83/0.77/0.88	0.59/0.56/0.61
Scalp: 0.75	0.39/0.37/0.40	0.85/0.80/0.89	0.57/0.54/0.60
Scalp: 0.8	0.37/0.34/0.39	0.87/0.82/0.90	0.56/0.53/0.59
Scalp: 0.85	0.35/0.32/0.38	0.88/0.84/0.91	0.55/0.52/0.59
Scalp: 0.9	0.34/0.30/0.37	0.90/0.86/0.93	0.55/0.50/0.59
Scalp: 0.95	0.36/0.31/0.41	0.92 /0.88/0.95	0.58/0.52/0.62
BBKNN	0.61/0.60/0.62	0.46/0.44/0.47	0.53/0.51/0.54
ComBat	0.47/0.39/0.53	0.61/0.63/0.59	0.53/0.50/0.56
Harmony	0.62/0.59/0.64	0.47/0.54/0.43	0.54/0.56/0.52
Scanorama	0.57/0.54/0.59	0.50/0.52/0.49	0.53/0.53/0.53

Table 2. Evaluating various methods via our method. Label and Batch columns show scores for label(do neighbors share cell types?) and batch based (do neighbors NOT share the same batch) score. In the last column we report the geometric mean over all relevant data. for each entry we report three groups: all data/Time series/batch-effect data.

Method	Label	Batch	Geomean
Scalp: 0.1	0.41 /0.34/0.47	0.89/0.87/0.91	0.61 /0.54/0.65
Scalp: 0.15	0.40/0.34/0.46	0.90 /0.88/0.92	0.60/0.54/0.65
Scalp: 0.2	0.41 /0.34/0.46	0.90 /0.88/0.92	0.61 /0.55/0.65
Scalp: 0.25	0.40/0.34/0.45	0.90 /0.88/0.92	0.60/0.55/0.64
Scalp: 0.3	0.40/0.33/0.45	0.90 /0.88/0.92	0.60/0.54/0.64
Scalp: 0.35	0.39/0.33/0.43	0.90 /0.88/0.92	0.59/0.54/0.63
Scalp: 0.4	0.38/0.33/0.41	0.90 /0.88/0.91	0.58/0.54/0.61
Scalp: 0.45	0.36/0.31/0.40	0.89/0.87/0.91	0.57/0.52/0.60
Scalp: 0.5	0.36/0.30/0.41	0.89/0.87/0.91	0.57/0.51/0.61
Scalp: 0.55	0.34/0.30/0.38	0.89/0.87/0.90	0.55/0.51/0.58
Scalp: 0.6	0.33/0.28/0.36	0.88/0.85/0.89	0.54/0.49/0.57
Scalp: 0.65	0.32/0.29/0.34	0.87/0.85/0.88	0.52/0.49/0.54
Scalp: 0.7	0.30/0.28/0.32	0.86/0.85/0.87	0.51/0.48/0.52
Scalp: 0.75	0.28/0.26/0.29	0.85/0.83/0.86	0.49/0.47/0.50
Scalp: 0.8	0.29/0.26/0.31	0.84/0.80/0.86	0.49/0.45/0.51
Scalp: 0.85	0.28/0.26/0.30	0.83/0.79/0.86	0.48/0.45/0.51
Scalp: 0.9	0.29/0.25/0.31	0.84/0.80/0.87	0.49/0.44/0.52
Scalp: 0.95	0.29/0.26/0.32	0.85/0.79/0.89	0.50/0.45/0.53
BBKNN	0.39/0.35/0.42	0.75/0.75/0.75	0.54/0.51/0.56
ComBat	0.37/0.30/0.42	0.87/0.83/0.90	0.57/0.50/0.62
Harmony	0.36/0.34/0.37	0.81/0.84/0.79	0.54/0.54/0.54
Scanorama	0.37/0.36/0.38	0.71/0.74/0.70	0.51/0.52/0.51

Table 3. Evaluating various methods via the SCIB method. Label and Batch columns show scores for label(do neighbors share cell types?) and batch based (do neighbors NOT share the same batch) score. In the last column we report the geometric mean over all relevant data. for each entry we report three groups: all data/Time series/batch-effect data.

	0
Harmony	0.53
Scanorama	0.51
BBKNN	0.56
ComBat	0.10
Scalp: 0.1	0.20
Scalp: 0.15	0.27
Scalp: 0.2	0.33
Scalp: 0.25	0.38
Scalp: 0.3	0.43
Scalp: 0.35	0.46
Scalp: 0.4	0.49
Scalp: 0.45	0.51
Scalp: 0.5	0.53
Scalp: 0.55	0.53
Scalp: 0.6	0.54
Scalp: 0.65	0.54
Scalp: 0.7	0.54
Scalp: 0.75	0.54
Scalp: 0.8	0.52
Scalp: 0.85	0.54
Scalp: 0.9	0.54
Scalp: 0.95	0.56
	0
Harmony	0.53
Scanorama	0.51
BBKNN	0.56
ComBat	0.10
Scalp: 0.1	0.21
Scalp: 0.15	0.29
Scalp: 0.2	0.35
Scalp: 0.25	0.41
Scalp: 0.3	0.46
Scalp: 0.35	0.50
Scalp: 0.4	0.53
Scalp: 0.45	0.55
Scalp: 0.5	0.57
Scalp: 0.55	0.58
Scalp: 0.6	0.58
Scalp: 0.65	0.59
Scalp: 0.7	0.59
Scalp: 0.75	0.59
Scalp: 0.8	0.59
Scalp: 0.85	0.58
Scalp: 0.9	0.58
Scalp: 0.95	0.57